

Binomial Expansion Solutions

1. Determine the first four terms of the Binomial Expansion $(3 + 2x)^6$:

Using the binomial expansion formula:

$$(a + b)^n = \sum (nCk) a^{(n-k)} b^k$$

Where:

- $a = 3$

- $b = 2x$

- $n = 6$

The first four terms:

$$= (6C0)(3^6)(2x)^0 + (6C1)(3^5)(2x)^1 + (6C2)(3^4)(2x)^2 + (6C3)(3^3)(2x)^3$$

$$= 1(729)(1) + 6(243)(2x) + 15(81)(4x^2) + 20(27)(8x^3)$$

$$= 729 + 2916x + 4860x^2 + 4320x^3$$

2. Find the value of $(0.31)^7$ by Binomial Expansion:

Let:

$$0.31 = (0.3 + 0.01)$$

Expand using the binomial theorem:

$$(0.3 + 0.01)^7 = \sum (7Ck) (0.3)^{(7-k)} (0.01)^k$$

First two terms:

$$= (7C0)(0.3^7)(0.01^0) + (7C1)(0.3^6)(0.01^1)$$

$$= 1(0.0002187) + 7(0.000729)(0.01)$$

$$= 0.0002187 + 0.00005103$$

Approximately 0.0002697

3. Determine the first term of the Binomial Expansion:

Given:

$$1 / (2x - 5y)^4$$

Rewrite as:

$$(2x - 5y)^{-4}$$

The first term:

$$= {}^4C_0 (2x)^{-4} (-5y)^0$$

$$= 1 * (2x)^{-4} * 1$$

$$= 1 / (16x^4)$$